Conceptual dependencies

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Conceptual Dependencies

- Conceptual Dependency theory is based on the use of a limited number of primitive concepts and rules of formation to represent any natural language statement.
- The theory states that different sentences which have the same meaning should have the same unique CD representation .
- Representation for any sentence should be unambiguous as the speaker intends, even though it may have syntactic ambiguity.
- Eg. “I saw the Golden Gate Bridge flying into San Francisco this afternoon”
- It is the contention that any concept, like dreaming, thinking, bellowing, or scheming can be described in terms of these primitives.

Canonical Meaning Representations

John gave Mary a book.

John gave a book to Mary.

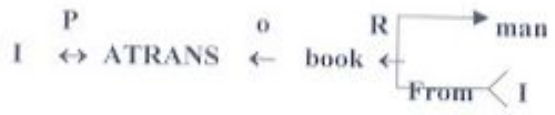
Mary was given a book by John.

Mary took a book from John.

Mary received a book from John.

I gave the man a book.

In CD from the above sentence can be represented with primitives as



I gave the man a book.

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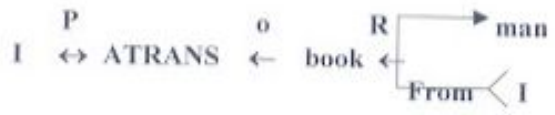


Conceptual Dependencies (Contd.)

- In CD theory five different types of ontological (state of being) building blocks are distinguished. Each of these types in turn has several subtypes.
- The types are made up of entities, actions, conceptual cases, conceptual dependencies, and conceptual tenses.

Entities

- Picture producers (PP) are actors or physical objects (including human memory) that perform different acts.
- Picture aiders (PA) are supporting properties or attributes of producers.

Actions

- Also know as primitive actions (ACTs).
- Action aiders (AA) are properties or attributes of primitive actions.

- ATRANS_ transfer of abstract entity
- ATTEND- focussing attention on an object
- CONC- to think about something
- GRASP- holding an object tightly
- MOVE-moving part of the body
- PTRANS-physical transfer from one location to another
- MTRANS_transfer of mental information
- INGEST- Ingesting something
- SPEAK-uttering sound

Conceptual Cases

- Objective Case
- Directive Case
- Instrumental Case
- Recipient Case

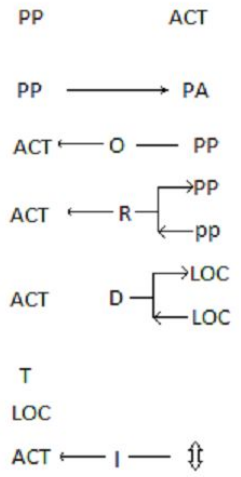
Conceptual Dependencies

- Semantic rules for the formation of dependency structures such as the relation between an actor and an event or between a primitive action and an instrum.

Conceptual dependancies (Contd.)

- Conceptual structures in the form of a graph are used to represent the meanings of different English (or other language) sentences.
- The graphs are constructed from elementary structures in accordance with basic syntax rules.
- Some of the basic concept rules are as follows:

Conceptual dependancies (Contd.)



Conceptual Tenses

- Time of action or state of being
- Conditional (c)
- Continuing (k)
- Finished Translation (tf)
- Future (f)
- Interrogative (?)
- Negative (/)
- Past (p)
- Present (nil)
- Start Transition (ts)
- Timeless (delta)
- Transition (t)

Conceptual dependancies (Contd.)

- Using these syntactic elements, structures which represent any sentence can be constructed.
- Complex sentence representations are constructed from the basic building blocks. Note the similarities between CD theory and the conceptual graphs of the previous section. Both have primitive concepts and relations defined, and both have a syntax for graphical representation. Conceptual graphs differ from CDs mainly in that the conceptual graph is logic based, whereas CD theory is mainly concerned with the semantics of events.

States

- states of objects are described by scales with numerical values.

HEALTH	(range -10 to 10)
dead	-10
gravely ill	-9
sick	-9 to -1
under the weather	-2
all right	0
tip top	+7
perfect health	+10

More States

FEAR	(range -10 to 0)
terrified	-9
scared	-5
anxious	-2
calm	0

MENTAL STATE	(range -10 to +10)
catatonic	-9
depressed	-5
upset	-3
sad	-2
ok	0
pleased	+2
happy	+5
ecstatic	+10

Combinations of States

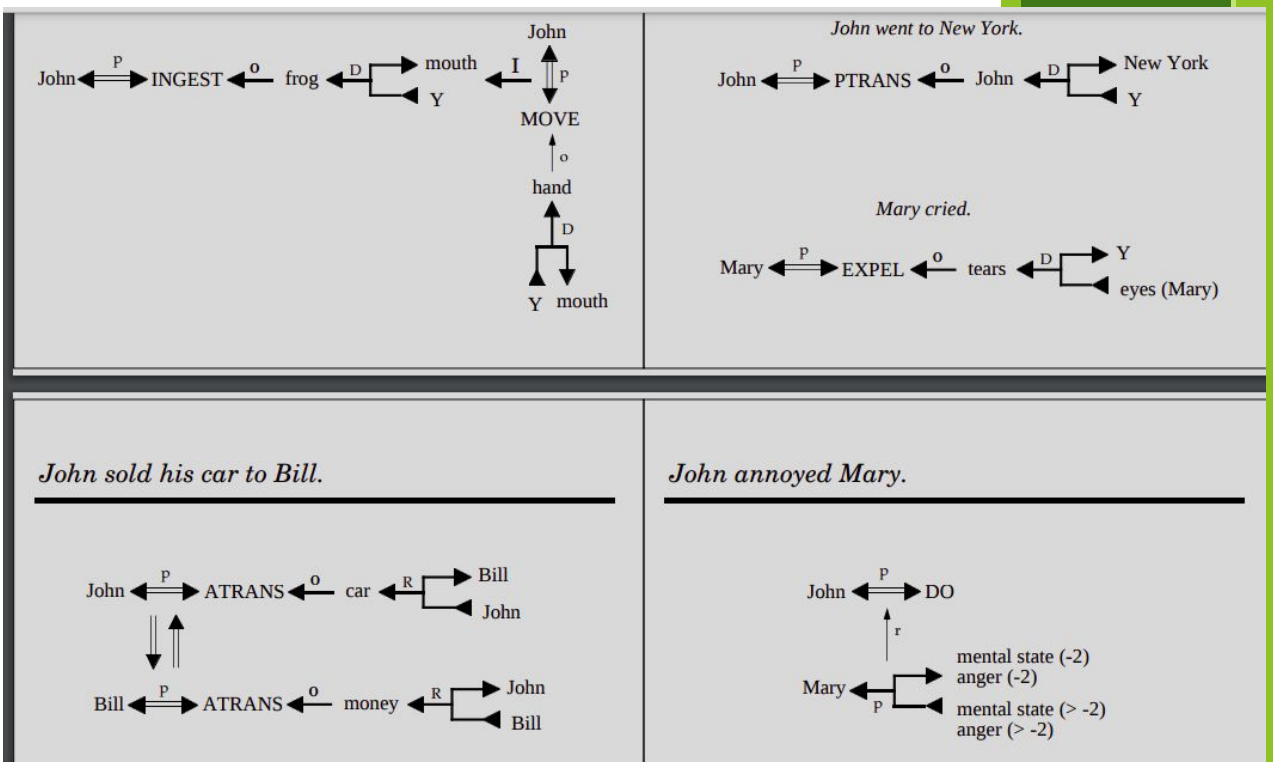
CONSCIOUSNESS	(range 0 to +10)
unconscious	0
asleep	5
awake	10
"higher drug consciousness"	>10

Some words can be combinations of scales:

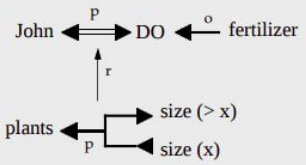
shocked = SURPRISE (6)
DISGUST (-5)

calm = SURPRISE (0)
DISGUST (0)
FEAR (0)
ANGER (0)
CONSCIOUSNESS (> 0)

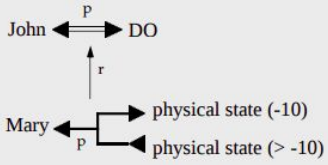
CD examples


John
P
INGEST
o
frog
D
mouth
Y
I
John
P
MOVE
o
hand
D
Y
mouth
John went to New York.
John
P
PTRANS
o
John
D
New York
Y
Mary cried.
Mary
P
EXPEL
o
tears
D
Y
eyes (Mary)
John sold his car to Bill.
John
P
ATRANS
o
car
R
Bill
John
Bill
P
ATRANS
o
money
R
John
Bill
John annoyed Mary.
John
P
DO
r
Mary
P
mental state (-2)
anger (-2)
mental state (> -2)
anger (> -2)

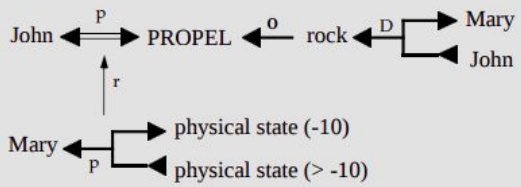
John grew the plants with fertilizer.



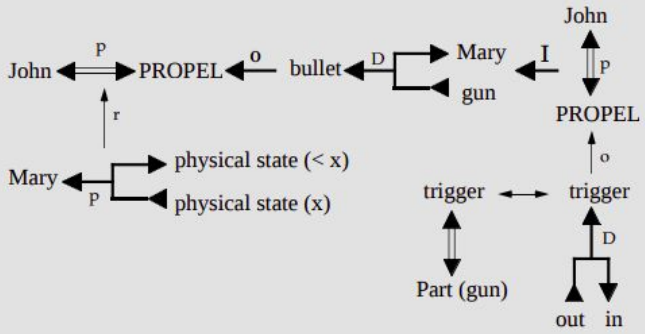
John killed Mary.



John killed Mary by throwing a rock at her.



John shot Mary.



• Weaknesses of CD

- **Incompleteness**

- no ontology (e.g., no ISA hierarchy)
- missing entire conceptual areas (e.g., objects)
- no method for handling quantification

- **The Primitives**

- are the primitives really atomic? (e.g., MOVE to a doctor

- More CD representation examples:
- visit the link:
- https://people.cs.pitt.edu/~wiebe/courses/CS3730/Spring2013/cd.pdf

The End